

DESCHAMBAULT, QC, Canada

WMO: 713890

Lat: 46.6906N

Lon: 71.9717W

Elev: 200

StdP: 14.59

Time Zone: -5.00 (NAE)

Period: 97-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-18.4	-13.2	-26.5	1.3	-16.8	-21.0	1.7	-11.7	18.5	16.9	16.7	17.1	2.7	320	0.545

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.5	84.5	70.6	81.9	68.9	79.1	67.4	73.8	80.9	71.8	78.6	69.8	76.0	5.1	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
71.5	117.3	78.3	69.5	109.1	75.2	67.6	102.2	73.4	37.5	80.6	35.6	78.8	34.0	76.0	82.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.2	14.1	12.3	DB	-27.6	89.6	5.1	2.7	-31.3	91.5	-34.3	93.1	-37.2	94.6	-40.9	96.6	(4)
(5)			WB	-27.4	77.6	5.1	2.4	-31.1	79.3	-34.1	80.7	-37.0	82.1	-40.7	83.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	40.3	10.6	13.9	24.7	38.7	52.5	60.8	65.9	64.1	56.2	44.1	32.3	18.4		
	DBStd	21.18	13.21	11.95	11.08	7.26	7.32	6.35	5.17	5.38	7.16	7.49	8.85	11.71		
	HDD50	5169	1222	1011	785	344	57	3	0	0	21	215	532	979		
	HDD65	9191	1687	1431	1250	789	394	157	51	83	276	649	980	1444		
	CDD50	1634	0	0	0	5	133	325	494	437	208	31	1	0		
	CDD65	183	0	0	0	0	5	30	80	55	13	0	0	0		
	CDH74	2136	0	0	0	4	145	418	876	540	151	2	0	0		
	CDH80	444	0	0	0	0	27	96	205	91	25	0	0	0		
	WSAvg	4.9	5.2	5.6	6.0	6.2	5.5	4.4	3.7	3.5	3.7	4.6	5.0	4.9		
	PrecAvg	48.1	3.4	2.8	2.9	3.3	3.7	4.4	5.6	4.6	4.8	4.6	3.8	4.1		
(15) Precipitation	PrecMax	55.4	5.6	4.6	5.2	7.6	6.1	8.2	9.4	6.9	9.1	9.2	6.2	8.9		
	PrecMin	38.9	1.5	1.1	0.8	0.7	2.0	2.0	2.8	1.5	2.4	1.6	1.4	1.8		
	PrecStd	4.8	1.2	0.9	1.0	1.7	1.1	1.7	1.7	1.2	1.7	1.7	1.1	1.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	40.8	41.7	53.7	70.1	83.7	86.8	89.6	85.6	83.5	71.8	57.7	43.0		
		MCWB	39.6	37.2	43.0	53.6	64.5	70.6	75.5	71.8	71.7	61.6	52.1	40.9		
	2%	DB	36.4	37.7	46.2	63.1	78.1	82.8	85.0	82.6	78.4	65.2	51.9	39.2		
		MCWB	34.8	34.1	38.0	49.4	61.0	67.8	71.9	70.2	68.2	57.1	47.9	37.5		
	5%	DB	33.5	35.0	42.4	57.0	73.0	79.0	82.1	79.8	74.2	60.2	47.8	36.0		
		MCWB	32.3	32.7	36.4	45.5	58.7	65.8	69.8	68.5	65.5	54.3	44.4	34.5		
	10%	DB	29.8	32.2	39.0	52.1	68.0	75.1	79.1	76.9	69.9	56.2	44.4	33.7		
		MCWB	28.3	30.3	34.5	42.6	55.9	63.3	68.3	67.2	63.0	51.5	41.6	32.5		
	0.4%	WB	39.9	38.4	45.4	58.1	69.7	74.2	77.3	75.5	73.7	63.7	54.7	41.9		
		MCDB	40.5	40.6	49.5	68.3	78.5	81.9	86.0	83.1	80.1	68.6	56.5	42.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	35.2	35.0	40.0	50.6	64.7	71.2	74.5	72.7	70.1	58.7	48.6	37.9		
		MCDB	36.0	36.8	44.3	58.6	72.8	78.9	81.8	79.2	75.5	63.1	50.3	39.0		
	5%	WB	32.5	33.0	37.3	47.1	60.7	68.5	72.2	70.7	67.1	55.5	44.7	34.9		
		MCDB	33.5	34.7	41.0	55.0	67.5	75.3	79.1	76.6	71.8	59.1	47.0	35.7		
	10%	WB	28.6	30.3	35.0	44.0	57.6	66.1	70.1	68.9	64.2	52.2	41.9	32.7		
		MCDB	29.7	32.3	38.4	50.2	66.4	72.5	76.6	74.5	68.9	55.2	44.0	33.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	19.7	21.7	20.1	20.3	23.7	23.5	23.5	23.4	23.4	18.6	14.7	16.0		
		MCDBR	16.0	17.5	22.4	31.6	33.4	29.7	27.7	26.7	27.7	25.4	19.4	12.6		
		MCWBR	15.9	14.7	15.5	19.3	18.3	16.8	15.9	15.5	17.7	17.8	15.8	11.8		
		MCDBR	15.8	15.6	19.0	26.8	27.0	24.9	24.2	22.9	23.9	22.7	16.8	12.3		
	5% WB	MCWBR	15.9	13.9	14.2	18.1	17.0	15.7	15.2	14.6	16.6	16.8	14.8	12.0		
(40) Clear-Sky Solar Irradiance	taub		0.265	0.283	0.320	0.348	0.366	0.407	0.411	0.399	0.371	0.337	0.308	0.284		
	taud		2.309	2.245	2.244	2.372	2.401	2.283	2.288	2.326	2.395	2.490	2.481	2.320		
	Ebn at Noon		264	281	286	288	285	273	270	268	266	261	248	240		
	Edh at Noon		24	31	37	36	36	41	40	37	32	25	21	21		
(44) All-Sky Solar Radiation	RadAvg		446	733	1133	1423	1616	1763	1729	1506	1180	689	402	301		
	RadStd		64	90	98	128	168	120	165	83	98	57	50	27		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (39 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page